

KING KHALID UNIVERSITY
COLLEGE OF COMPUTER SCIENCE

INVENTORY MANAGEMENT SYSTEM

By

Name	Academic Number	Role
Renad Mohammed Alshahri	444xxxxxx	Project Manager
Wajd Yahya Asiri	444xxxxxx	Risk Manager & Documentation Specialist

Course Code: 444CCS-3
Course Name: Software Project Management
Instructor: Dr.
2nd Semester – AY 1447

Abstract

This report presents the project management plan for the development of an Inventory Management System (IMS). The system is designed to automate and streamline inventory tracking, stock control, and reporting processes for a retail or warehousing environment. This document covers all phases of the project lifecycle, from planning and organization through risk management and conclusion. It applies standard software project management methodologies including scope definition, work breakdown structure, Gantt scheduling, cost estimation, risk identification, and communication planning. The project is carried out by a team of two students as part of the 444CCS-3 Software Project Management course at King Khalid University.

Table of Contents

Chapter 1: Project Introduction **6**
1.1 Project Overview 6
1.2 Project Scope 6
1.3 Team Structure 7
1.4 Key Stakeholders 7

Chapter 2: Project Planning and Organization **8**
2.1 Software Project Lifecycle Model 8
2.2 Work Breakdown Structure (WBS) 9
2.3 Project Schedule 10
2.4 Resource Allocation 11

Chapter 3: Project Management Plan **12**
3.1 Scope Management 12
3.2 Time Management 12
3.3 Cost Management 12
3.4 Communication Management 13

Chapter 4: Risk Management **14**
4.1 Risk Identification & Analysis 14
4.2 Risk Response Planning 14
4.3 Monitoring and Reporting 15

Chapter 5: Conclusion **16**

References **17**

List of Figures

Figure 1: Agile Scrum Lifecycle Model 8
Figure 2:Work Breakdown Structure (WBS) 9
Figure 3:Gantt Chart Schedule ..... **خطأ! الإشارة المرجعية غير معرّفة.**

List of Tables

Table 1: Team Structure 7
Table 2:Key Stakeholders 7
Table 3:Project Schedule 10
Table 4:Resource Allocation 11
Table 5:Cost Management 13
Table 6:Risk Identification & Analysis 14
Table 7:Risk Response Planning 14

Chapter 1: Project Introduction

1.1 Project Overview

The Inventory Management System (IMS) is a software solution designed to help businesses efficiently manage their inventory, track stock levels, process orders, and generate detailed reports. The system eliminates manual inventory tracking processes that are prone to errors and inefficiencies.

Objectives:

- Automate the tracking of inventory items across multiple storage locations.
- Provide real-time stock level visibility to authorized users.
- Enable efficient order processing, restocking alerts, and supplier management.
- Generate comprehensive inventory reports for decision-making.
- Reduce inventory costs by minimizing overstocking and stockouts.

Justification: Manual inventory management leads to errors, delays, and financial losses. An automated IMS will improve accuracy, reduce operational costs, and enhance decision-making capabilities for the organization.

1.2 Project Scope

In Scope:

- User authentication and role-based access control (Admin, Manager, Staff).
- Product catalog management (add, edit, delete, categorize items).
- Stock level tracking with automated low-stock alerts.
- Purchase order management and supplier database.
- Sales order processing and invoice generation.
- Reporting module: stock reports, sales reports, audit trails.
- Web-based interface accessible from desktop browsers.

Out of Scope:

- Mobile application development.
- Integration with third-party ERP systems.
- Hardware barcode scanner integration.

Success Criteria:

- System deployed and functional within the 12-week timeline.
- All core modules tested and passing acceptance criteria.
- User manual and technical documentation delivered.
- Project delivered within the estimated budget.

1.3 Team Structure

Table 1: Team Structure

Role	Responsibilities	Student
Project Manager	Overall planning, coordination, scheduling, tracking progress, leading team meetings, scope and time management.	Renad Mohammed Alshahri
Risk Manager & Documentation Specialist	Risk identification, risk log maintenance, mitigation strategies, report writing, stakeholder communication, deliverable compilation.	Wajd Yahya Asiri

1.4 Key Stakeholders

Table 2:Key Stakeholders

Stakeholder	Type	Interest / Role
Project Team	Internal	Development and management of the project.
Course Instructor	Internal	Academic supervisor and evaluator.
End Users (Staff/Managers)	External	Will use the system daily for inventory operations.
Business Owner / Client	External	Requires the system for business operations.
Suppliers	External	Indirectly affected through purchase order processing.

Chapter 2: Project Planning and Organization

2.1 Software Project Lifecycle Model

The project will adopt an Agile (Scrum) lifecycle model with elements of traditional planning. Agile is chosen because the requirements may evolve during development, and iterative delivery allows for early feedback and course correction.

Justification for Agile:

- Flexibility to accommodate changing requirements.
- Iterative sprints (2-week cycles) ensure frequent deliverables.
- Regular stakeholder feedback improves product quality.
- Clear roles align well with our two-member team structure.

Sprint Plan: The project is divided into 6 sprints of approximately 2 weeks each, covering requirements analysis, design, development of each module, testing, and deployment.

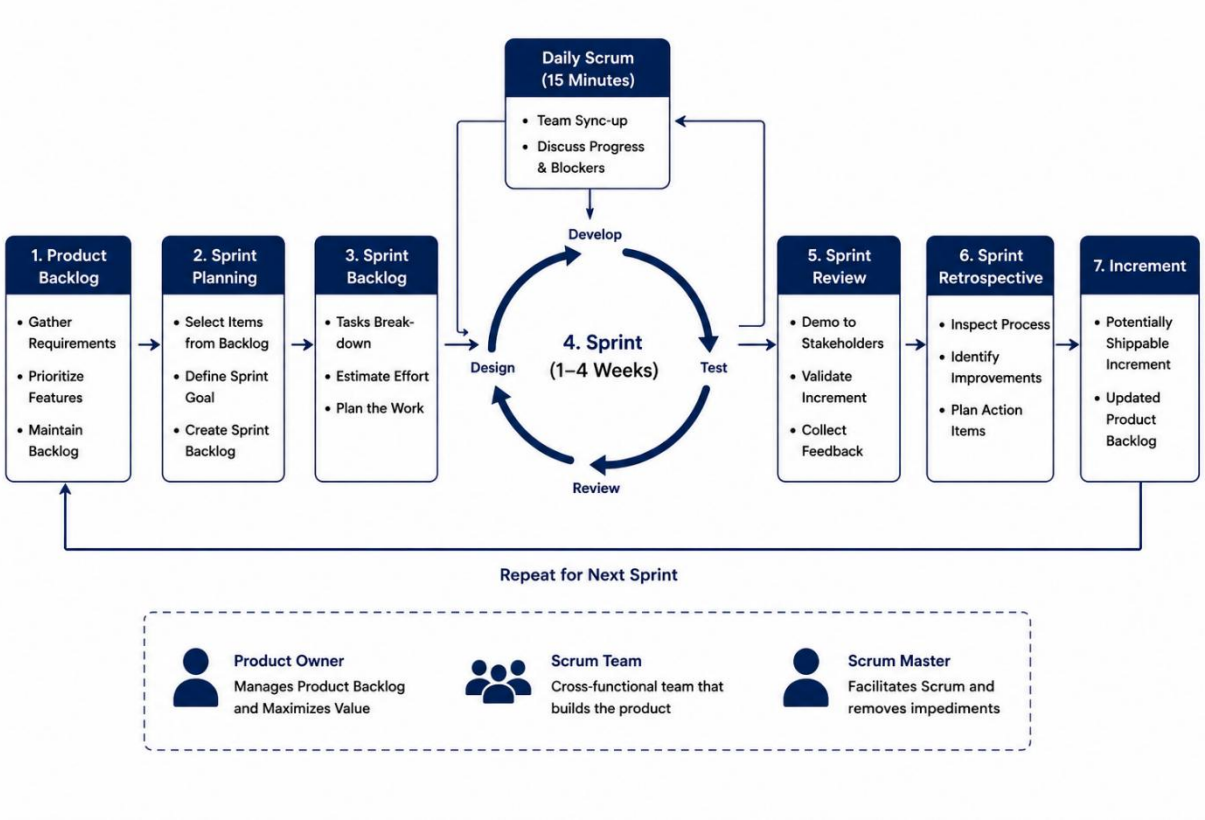


Figure 1: Agile Scrum Lifecycle Model

2.2 Work Breakdown Structure (WBS)

The WBS hierarchically decomposes the project into manageable tasks:

- 1. Project Management: 1.1 Planning, 1.2 Monitoring, 1.3 Reporting
- 2. Requirements: 2.1 Stakeholder Interviews, 2.2 Requirements Document, 2.3 Approval
- 3. System Design: 3.1 Architecture Design, 3.2 Database Design, 3.3 UI/UX Wireframes
- 4. Development: 4.1 Authentication Module, 4.2 Product Module, 4.3 Inventory Module, 4.4 Order Module, 4.5 Reporting Module
- 5. Testing: 5.1 Unit Testing, 5.2 Integration Testing, 5.3 User Acceptance Testing
- 6. Deployment: 6.1 Server Setup, 6.2 System Deployment, 6.3 User Training
- 7. Documentation: 7.1 Technical Manual, 7.2 User Manual, 7.3 Final Report

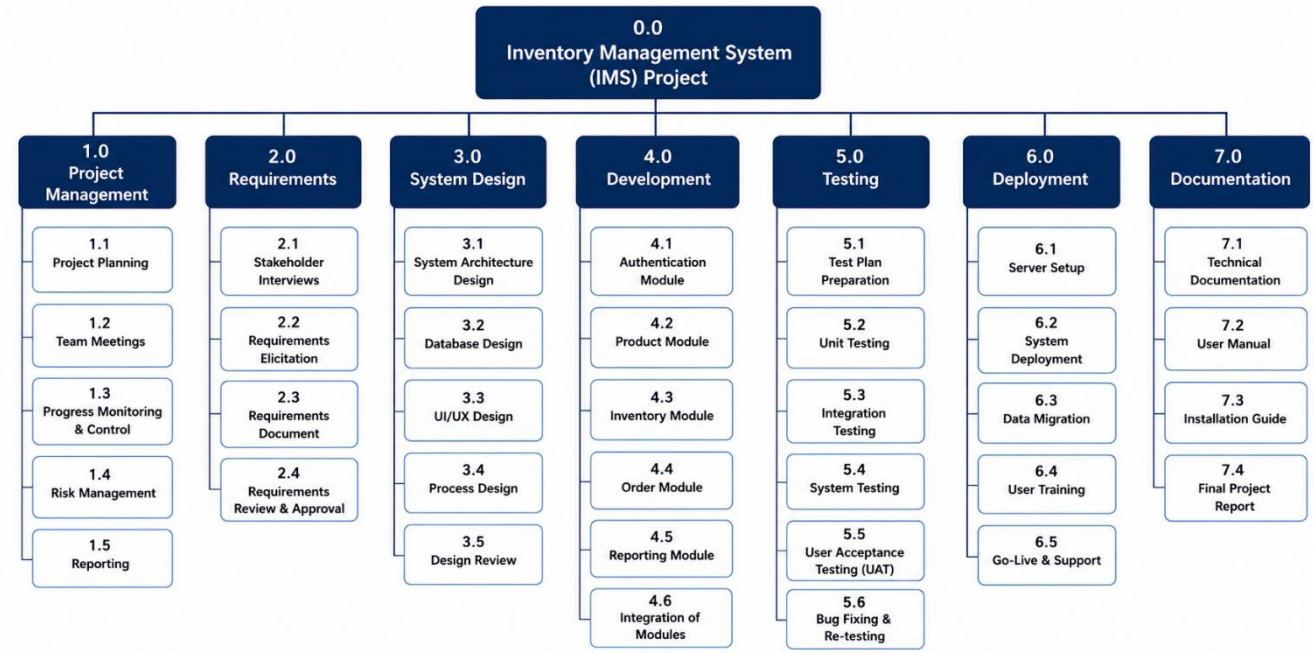


Figure 2:Work Breakdown Structure (WBS)

2.3 Project Schedule

The project spans 12 weeks. Key milestones:

Table 3:Project Schedule

Week	Phase / Task	Milestone	Responsible
1-2	Requirements Gathering & Analysis	Requirements Document Approved	Renad Mohammed Alshahri
3-4	System Design (Architecture, DB, UI)	Design Documents Complete	Both
5-6	Development: Auth & Product Modules	Sprint 1 Review	Both
7-8	Development: Inventory & Order Modules	Sprint 2 Review	Both
9	Development: Reporting Module	Sprint 3 Review	Both
10	Testing (Unit, Integration, UAT)	Testing Complete	Both
11	Deployment & User Training	System Live	Renad Mohammed Alshahri
12	Final Documentation & Report Submission	Project Closure	Wajd Yahya Asiri

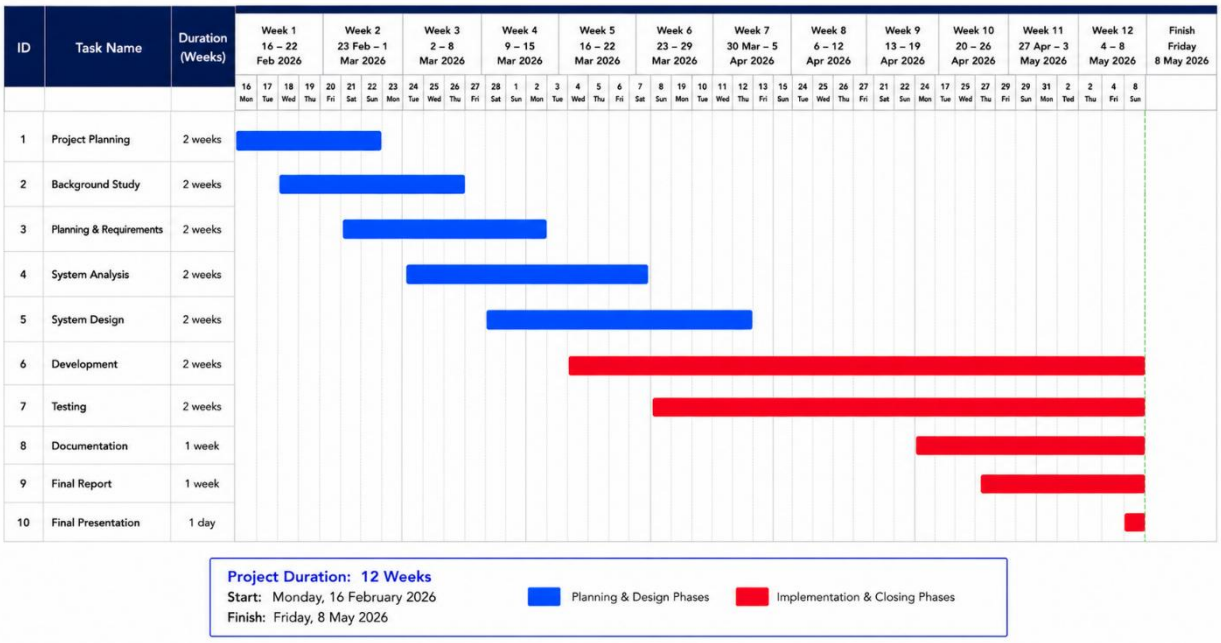


Figure 3: Gantt Chart Schedule

2.4 Resource Allocation

Table 4:Resource Allocation

Resource	Type	Role in Project	Availability
Renad Mohammed Alshahri	Human	Project Manager	Full project duration
Wajd Yahya Asiri	Human	Risk Manager & Doc Specialist	Full project duration
Visual Studio Code	Technical	Development IDE	Free – All sprints
GitHub	Technical	Version Control	Free – Full project
MySQL	Technical	Database Management	Free – Development phase
GanttPro	Technical	Project Scheduling	Free trial
Microsoft Teams	Technical	Team Communication	University license
Blackboard	Technical	Submission Platform	University provided

Chapter 3: Project Management Plan

3.1 Scope Management

Scope management ensures the project includes all required work and only the required work.

Requirements Definition Process:

- Conduct stakeholder interviews to elicit requirements.
- Document functional and non-functional requirements in a Software Requirements Specification (SRS).
- Review and obtain formal approval from the instructor/client before development begins.

Scope Change Handling:

- Any scope change request must be submitted in writing using a Change Request Form.
- The project team evaluates impact on schedule, cost, and quality.
- Changes approved by the Project Manager are incorporated into updated project documents.
- A change log is maintained throughout the project.

3.2 Time Management

Time management ensures the project meets its 12-week deadline.

Detailed Schedule: Gantt chart maintained in GanttPro showing all tasks, durations, dependencies, and milestones (see Section 2.3).

Milestone Tracking:

- Weekly progress reviews every Sunday to assess milestone completion.
- Status reported in weekly meeting minutes.
- Any delayed tasks flagged immediately for corrective action.

Buffer Management:

- Week 10 allocated for testing serves as a natural buffer.
- Each sprint includes a 1-day buffer before the review meeting.
- Critical path activities monitored closely to avoid deadline slippage.

3.3 Cost Management

Since this is an academic project, the cost is primarily measured in effort (hours). Estimated effort for two team members:

Table 5:Cost Management

Phase	Estimated Hours	Cost Equivalent (SAR 50/hr)	Notes
Requirements Analysis	16 hrs	SAR 800	Team effort
Design	20 hrs	SAR 1,000	Team effort
Development	70 hrs	SAR 3,500	Main development phase
Testing	16 hrs	SAR 800	Testing phase
Documentation	14 hrs	SAR 700	Reports and manuals
Deployment	6 hrs	SAR 300	Setup and training
Contingency Reserve	14 hrs	SAR 700	10% buffer
Total	156 hrs	SAR 7,800	Full project

Tools used are either free (open-source) or available through university licensing, minimizing external costs.

3.4 Communication Management

Reporting Structure: Project Manager reports progress to the Instructor. Both team members report mutually and maintain shared documentation.

Meeting Calendar:

- Weekly team meetings: Every Sunday at 7:00 PM via Microsoft Teams.
- Sprint Review meetings: At the end of each 2-week sprint.
- Ad-hoc meetings: As needed for critical issues.

Documentation Tools:

- Microsoft Word: For all formal reports and documentation.
- GitHub Wiki: For technical documentation and code documentation.
- Microsoft Teams: For daily communication and file sharing.
- Blackboard: For formal submission of deliverables.
- Meeting minutes documented and distributed within 24 hours of each meeting.

Chapter 4: Risk Management

4.1 Risk Identification & Analysis

The following risk log identifies potential risks with their probability (H/M/L) and impact (H/M/L) assessments:

Table 6:Risk Identification & Analysis

Risk ID	Risk Description	Probability	Impact	Risk Level
R01	Team member becomes unavailable due to illness or personal issues	Low	High	Medium
R02	Requirement changes after development begins	Medium	High	High
R03	Technical difficulties with development tools/environment	Medium	Medium	Medium
R04	Underestimation of task complexity leading to schedule overrun	Medium	High	High
R05	Data loss due to lack of version control discipline	Low	High	Medium
R06	Security vulnerabilities in authentication module	Medium	High	High
R07	Inadequate testing leading to critical bugs at delivery	Low	High	Medium
R08	Workload imbalance between the two team members	Medium	Medium	Medium

4.2 Risk Response Planning

Table 7:Risk Response Planning

Risk ID	Strategy	Response Action	Owner
R01	Mitigation	Document all work clearly on GitHub so the other member can continue. Cross-train on all modules.	Both
R02	Avoidance	Lock requirements after SRS approval. Use formal Change Request process for any changes.	Renad Mohammed Alshahri
R03	Mitigation	Set up development environments in week 1. Use standardized tech stack. Document setup steps.	Renad Mohammed Alshahri
R04	Mitigation	Include buffer time in schedule. Review WBS estimates before finalizing. Break tasks into smaller units.	Renad Mohammed Alshahri

R05	Mitigation	Use GitHub with mandatory commits after each session. Protect main branch with pull requests.	Both

R06	Mitigation	Implement secure password hashing (bcrypt). Conduct security review of auth module before testing.	Wajd Yahya Asiri
R07	Mitigation	Prepare comprehensive test cases. Allocate dedicated testing week. Conduct UAT with simulated users.	Wajd Yahya Asiri
R08	Avoidance	Define clear task ownership in WBS. Review workload distribution at every weekly meeting.	Both

4.3 Monitoring and Reporting

Risk monitoring is an ongoing activity throughout the project lifecycle:

- Risk log reviewed and updated at every weekly team meeting.
- Risk status update included at each sprint review session.
- New risks identified during development are added to the risk log immediately.
- Risks that materialize are addressed by both team members immediately.
- A risk summary section is included in every weekly progress report.
- Risk log is maintained in a shared document accessible to both team members via GitHub.

Chapter 5: Conclusion

Summary of What Was Achieved

This project management report has systematically defined and planned the development of an Inventory Management System (IMS). The following was accomplished:

- A comprehensive project scope with clear boundaries, deliverables, and success criteria was established.
- An Agile (Scrum) lifecycle model was selected and justified as the most appropriate approach.
- A detailed Work Breakdown Structure was developed covering all project phases.
- A 12-week Gantt schedule with milestones was created and documented.
- Resource allocation was planned for all human and technical resources.
- A complete project management plan covering scope, time, cost, and communication was developed.
- A thorough risk register with 8 identified risks, probability/impact assessments, and response strategies was created.

Key Learning Outcomes

- Applied project management methodologies (Agile Scrum) in a realistic software project context.
- Developed skills in scope definition, WBS creation, and Gantt chart scheduling.
- Gained experience in risk identification, analysis, and response planning.
- Understood the importance of communication planning and documentation in project success.
- Learned to estimate effort and manage resources effectively in a two-member team environment.
- Practiced team collaboration, role assignment, and accountability in a group project setting.

Possible Future Improvements / Extensions

- Develop a mobile application (iOS/Android) for on-the-go inventory management.
- Integrate with popular ERP systems such as SAP or Oracle for enterprise-level operations.
- Implement barcode and QR code scanning for rapid product entry and stock counting.
- Add machine learning-based demand forecasting to optimize reorder levels automatically.
- Expand the system to support multi-warehouse and multi-location inventory management.
- Implement a customer-facing portal for order tracking and delivery status.

References

Project Management Institute. (2021). A Guide to the Project Management Body of Knowledge (PMBOK Guide) (7th ed.). PMI.

Schwaber, K., & Sutherland, J. (2020). The Scrum Guide. Scrum.org. Retrieved from https://www.scrumguides.org

Sommerville, I. (2016). Software Engineering (10th ed.). Pearson.

Pressman, R. S., & Maxim, B. R. (2019). Software Engineering: A Practitioner's Approach (9th ed.). McGraw-Hill.

Kerzner, H. (2022). Project Management: A Systems Approach to Planning, Scheduling, and Controlling (13th ed.). Wiley.